MILITARY SPECIFICATION

BODY ASSEMBLY: 8619880

1. SCOPE

1.1 This specification covers a modular unit of the Tank Commander's M34 Periscope (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

F5549108	Telescope, Collimating (3X)
F8298789	Projector, Collimator, Parabolic
F8565556	Pressure Tester, Two-Station
F8566516	Adapter

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F8575858	Fixture, Body Assembly, Periscopes M34 and M36
F8577787	Universal Test Stand
C8591107	Adapter
C8591109	Adapter
F8619880	Body Assembly: 8619880
A10531935	Set-up Instructions for Test Fixture F8575858
A10531936	Operating Instructions for Test Fixture F8575858
F11732113	Gage, Interchangeability Maximum
F11732114	Gage, Interchangeability Minimum

PACKAGING DATA SHEET

8619880	Packaging of Body Assembly: 8619880

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- The materials shall be in accordance with the drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication.- The body assembly shall be manufactured in accordance with Drawing F8619880 and drawings pertaining thereto.

3.2.1 Interchangeability.- The body assembly shall be capable of insertion into and removal from the gages specified in 4.6.7.18.

3.3 General specifications.- The contractor shall be responsible for adherence to, and compliance with the following requirements of MIL-F-13926.

3.3.1 MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

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3.4 Environmental.

3.4.1 Storage temperature.- The body assembly shall show no evidence of cement separation, glass breakage or other physical failure, and shall meet all the requirements of this specification at standard ambient temperature (plus 60° to 90°F) after having been exposed and thermally stabilized at ambient temperatures of minus 80° to plus 160° Fahrenheit (F).

3.4.2 Operating temperatures.- The body assembly shall meet the applicable requirements of 3.7.1.13.1, 3.7.1.13.2, 3.7.1.13.3 and 3.7.1.13.4 while exposed and thermally stabilized at ambient temperatures of minus 40°F and plus 150°F, subsequent to which the body assembly shall meet the requirements of 3.4.3 at standard ambient temperatures (60° to 90°F).

3.4.3 Vibration.- The body assembly shall show no evidence of physical failure due to being vibrated in a vertical plane at a constant frequency of 30 cycles per second (CPS) with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes plus or minus 15 seconds. Following the vibration test, the body assembly shall meet all remaining requirements herein.

3.5 Cleanliness.

a. Defect.- There shall be no evidence of moisture, grease, fingerprints, condensates, fractures and adhesive separations on any glass components when viewing through either end (including the unity power system).

b. Dirt.- When viewed through either eyepiece there shall be no more than three dirt particles appearing on the reticle (or reticle blank) within a 50 mil diameter central field of view. There shall be no more than three dirt particles appearing outside the above field of view. The angular separation between dirt particles shall be at least 15 mils. Dirt particles smaller than 1/4 reticle line width shall be ignored regardless of distribution. There shall be no more than three particles of dirt or foreign material (classed as dirt) on any glass component of the unity power system and the window and objective lenses of the 7X system. The size of any dirt particle shall not exceed 0.05 inch.

3.6 Sealing.- The body assembly shall show no evidence of leakage in excess of 0.2 pound per square inch gage (psig) when subjected to an internal pressure of 5 ± 0.10 psig for a minimum of 2 hours. The interior atmosphere of the accepted body assembly shall contain nitrogen (having a dew point at least as low as minus 25°F at atmospheric pressure.) between 0.25 and 0.50 pound per square inch gage (psig).

3.7 Performance.

3.7.1 Seven power (7X) system.

NOTE: The following references apply throughout this specification:

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Datum Plane "X" is established by the Mounting Pins 8619855 and 8619903 in the top of the body assembly.
Datum Plane "Y" is perpendicular to Datum Plane "X" and passes through a line which is rotated counterclockwise 21 degrees 10 minutes from the centers of the two 0.1875 diameter dowel holes of Pins 8619903.
Elevation is measured perpendicular to Datum Plane "Y".
Deflection is measured parallel to Datum Plane "Y".

3.7.1.1 Parallax.- Parallax between the reticle and the image of a target at 1200 meters as viewed through the right eyepiece shall not exceed 0.1 mils.

3.7.1.2 Collimation.- These requirements shall be met prior to meeting the requirements of 3.4.3.

3.7.1.2.1 Previbration setting.- An incident line of sight shall be provided which is perpendicular to Datum Plane "X", within 0.2 mil. Coincidence of this line of sight with the instrument reticle shall be established by rotation of the boresight knobs.

3.7.1.2.2 Reticle movement.- Reticle (boresight cross) movement shall be measured against an external target from the collimated setting of 3.7.1.2.1 to meet the requirements of 3.7.1.2.2.1 and 3.7.1.2.2.2.

3.7.1.2.2.1 Elevation.- Clockwise rotation of the elevation knob shall move the reticle, in a direction towards the knob, a minimum of 8.5 mils from the collimated setting. Counterclockwise rotation of the elevation knob shall move the reticle, in a direction away from the knob, a minimum of 5.5 mils from the collimated setting.

3.7.1.2.2.2 Deflection.- Clockwise rotation of the deflection knob shall move the reticle, in a direction towards the knob, a minimum of 7.0 mils from the collimated setting. Counterclockwise rotation of the deflection knob shall move the reticle, in a direction away from the knob, a minimum of 7.0 mils from the collimated setting.

3.7.1.2.3 Knob rotation.- Continuous movement of the reticle shall be evident until limited by positive engagement of clockwise and counterclockwise stops in each boresight knob.

3.7.1.3 Line of sight variation.- Variation in the line of sight in elevation, as the deflection knob is rotated by at least 7.0 mils in each direction from the collimated setting of 3.7.1.2.1, shall not exceed 0.1 mil (total variation not to exceed 0.2 mil).

3.7.1.4 Reticle shift.- Disengagement or reversal of either knob shall not cause the line of sight to shift by more than 0.1 mil in a direction perpendicular to the intended motion.

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3.7.1.5 Boresight backlash.- Backlash in each boresight mechanism shall not exceed 0.1 mil.

3.7.1.6 Post vibration.- The original setting of each reticle boresight knob relative to the incoming line of sight established in 3.7.1.2.1 shall not deviate more than 0.1 mil (including fixture error) as observed through eyepiece subsequent to meeting requirements of 3.4.3.

3.7.1.7 Image tilt.- A line originally parallel to the intersection of Datum Planes "X" and "Y" shall emerge parallel to Datum Plane "X" within one (1) degree when measured through either eyepiece.

3.7.1.8 Parallelism of reticle and image.- A line, originally parallel to intersection of Datum Planes "X" and "Y" shall be parallel to the horizontal reticle line within 30 minutes of arc.

3.7.1.9 Dioptral adjustment.

3.7.1.9.1 Zero setting.- This requirement shall be met while observing the reticle through a dioptometer set for zero diopters. The diopter scale shall indicate zero within 1/4 diopter when the reticle is brought into focus. An image of a target at 1200 meters shall be used as the zero diopter reference for the left side.

3.7.1.9.2 Excursion range.- The diopter movement shall be adjustable through a minimum range of -3.0 to + 3.0 diopters as read on the diopter scale.

3.7.1.9.3 Diopter backlash.- Backlash in the diopter movement shall not exceed 1/8 diopter as read on the diopter scale.

3.7.1.10 Resolution.- The resolution shall be 8.0 seconds of arc or less when viewing through either eyepiece with an auxiliary telescope having a magnification of at least 3.0 power. Resolution consists of detectable line structure and proper line count in all four meridians. Eyepiece focus shift between any two meridians shall not exceed 0.25 diopters.

3.7.1.11 Binocular collimation.- Two parallel rays projected into the binocular 7X system shall emerge from the eyepieces with a divergence between 5 and 30 minutes. Dipvergence shall not exceed 15 minutes of arc. This requirement shall be met at the zero diopter setting of the eyepieces at any setting of the interpupillary scale between 58 and 72 millimeters.

3.7.1.12 Magnification.- The magnification of the two power systems shall not differ from each other by more than 2 percent of the larger magnification.

3.7.1.13 Operability.

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3.7.1.13.1 Diopter movement.- The running torque required to rotate either of the diopter movements through the range specified in 3.7.1.9.2 shall be within the limits of 1.5 and 6.0 lb-ins when checked at standard temperature. When checked at the temperature conditions specified in 3.4.2, the running torque shall not exceed 15 lb-ins.

3.7.1.13.2 Disengagement of boresight knob locks.- The direct force required to disengage each boresight knob from the lock position shall be within the limits of 2.0 to 5.0 lbs when checked at standard temperature. When checked at the temperature conditions specified in 3.4.2, the direct force shall not exceed 9.0 lbs.

3.7.1.13.3 Rotation of boresight knobs.- The running torque required to rotate each boresight knob through the range specified in 3.7.1.2.2 and 3.7.1.2.3 shall be within the limits of 0.5 and 5.0 lb-ins when checked at standard temperature. When checked at the temperature conditions specified in 3.4.2 the running torque shall not exceed 8.0 lb-ins.

3.7.1.13.4 Interpupillary distance knob torque.- The IPD knob torque shall not exceed 7 pound-inches at standard ambient temperature or 10 pound-inches at extreme temperatures as specified in 3.4.2.

3.7.1.14 Reticle illumination.- The reticle markings illuminated by control light source specified in 4.6.7.17 and viewed in a dark room shall appear clearly defined.

3.7.2 Unity power system.

3.7.2.1 Collimation.- A line of sight, originally perpendicular to Datum Plane "X", shall be deviated 20 degrees ± 30 minutes below the horizontal (in a plane perpendicular to Datum Planes "X" and "Y"). The deflection shall not exceed 30 minutes relative to the plane defined above.

3.7.2.2 Spherical power and astigmatism.- The spherical power and astigmatism shall not exceed 0.1 diopter when scanning the clear aperture with a dioptometer of at least 0.5 inch aperture.

3.7.2.3 Image tilt.- A line, originally parallel to the intersection of Datum Planes "Y" and "X", shall emerge parallel to Datum Plane "X" within 30 minutes of arc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three body assemblies plus three each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. In addition to 100% inspection as specified in 4.3.2.2, the tabulated classification of defects in Tables I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

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4.3.2.2 100 percent inspection.- The following requirements and related tests shall be subjected to 100% inspection:

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
1. Line of sight variation	3.7.1.3	4.6.7.5
2. Reticle shift	3.7.1.4	4.6.7.6
3. Dioptral adjustment, zero setting	3.7.1.9.1	4.6.7.10.1
4. Resolution	3.7.1.10	4.6.7.11
5. Reticle illumination	3.7.1.14	4.6.7.17
6. Interchangeability	3.2.1	4.6.7.18

NOTE: The tests in 4.3.2.2 shall be conducted at a temperature between + 60° and + 90°F.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 0.65% DEFECTIVE		
101. Vibration	3.4.3	4.6.4
102. Collimation, post vibration	3.7.1.6	4.6.7.1
103. Cleanliness	3.5	4.6.5
104. Sealing	3.6	4.6.6
105. Parallax	3.7.1.1	4.6.7.2
106. Collimation, pre-vibration setting	3.7.1.2.1	4.6.3
107. Reticle movement, elevation	3.7.1.2.2.1	4.6.7.3
108. Reticle movement, deflection	3.7.1.2.2.2	4.6.7.3.1
109. Knob rotation	3.7.1.2.3	4.6.7.4
110. Boresight backlash	3.7.1.5	4.6.7.7
111. Image tilt, seven power system	3.7.1.7	4.6.7.8
112. Parallelism of reticle and image	3.7.1.8	4.6.7.9
113. Dioptral adjustment, excursion range	3.7.1.9.2	4.6.7.10.2
114. Diopter backlash	3.7.1.9.3	4.6.7.10.3
115. Binocular collimation, dipvergence	3.7.1.11	4.6.7.12
116. Binocular collimation, divergence	3.7.1.11	4.6.7.13
117. Magnification	3.7.1.12	4.6.7.14
118. Collimation, unity power system	3.7.2.1	4.6.7.15.1
119. Image tilt, unity power system	3.7.2.3	4.6.7.15.2
120. Spherical power & astigmatism, unity power system	3.7.2.2	4.6.7.15.3
121. Operability, diopter movement	3.7.1.13.1	4.6.7.16.1
122. Operability, disengagement of boresight knob locks	3.7.1.13.2	4.6.7.16.2
123. Operability, rotation of boresight knobs	3.7.1.13.3	4.6.7.16.3
124. Operability, interpupillary distance knob	3.7.1.13.4	4.6.7.16.4
MINOR: NONE		

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NOTE: The tests in Table I shall be conducted at temperature between + 60° and + 90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One body assembly shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.2	Applicable Drawing - Visual
302.	General specifications	3.3	MIL-F-13926 - MIL-O-13830 - Visual

NOTE: The examinations in Table II shall be conducted at a temperature between 60° and 90°F.

4.4.2 Environmental.- Three body assemblies shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III and shall then meet the requirements and tests in Table I.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperatures (-80° and +160°F)	3.4.1	4.6.1
304.	Operability, diopter movement	3.4.2, 3.7.1.13.1	4.6.2, 4.6.7.16.1
305.	Operability, disengagement of boresight knob locks	3.4.2, 3.7.1.13.2	4.6.2, 4.6.7.16.2
306.	Operability, rotation of boresight knobs	3.4.2, 3.7.1.13.3	4.6.2, 4.6.7.16.3
307.	Operability, interpupillary distance knob	3.4.2, 3.7.1.13.4	4.6.2, 4.6.7.16.4

NOTE: The tests for Characteristics 304 through 307 inclusive shall be conducted at -40° and +150°F (see 3.4.2).

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4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted items have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- All inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Sealing.- Special testing equipment to test sealing of the body assembly shall conform to the internal pressure test requirements of Drawing F8565556, Pressure Tester-Two Station.

4.5.2.3.2 Performance.- Special testing equipment to test performance of the body assembly shall conform to the requirements of Drawing F8575858, Fixture, Body Assembly, Periscopes, M34 and M36; Drawing F8577787, Universal Test Stand and Drawing F8298789, Projector, Collimating, Parabolic.

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4.6 Test methods and procedures.

4.6.1 Storage temperatures.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.1. The body assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see NOTE below) to -80°F. The body assembly shall remain at this temperature for a minimum of 5 hours At the completion of this 5 hour period, the temperature of the test chamber shall be gradually raised (see NOTE below) to +160°F. The body assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, the temperature of the test chamber shall be gradually reduced to room ambient temperature (+60° to +90°F), and the body assembly removed from the test chamber. The body assembly shall then be subjected to the visual, tactile, and audible examination of 4.6.4. The body assembly shall conform to the requirements of 3.4.1.

NOTE: The rate of temperature change in the test chamber shall not exceed 20 degrees per hour throughout the temperature cycling tests of 4.6.1 and 4.6.2.

4.6.2 Operating temperatures.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.2. The body assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see NOTE in 4.6.1) to -40°F. The body assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period and while at 40°F, the body assembly shall be subjected to the tests in 4.6.7.16.1 to 4.6.7.16.4 inclusive, to determine compliance with the applicable requirements of 3.7.1.13.1 to 3.7.1.13.4 inclusive, for this temperature (-40°F). Upon completion of this test, the temperature of the test chamber shall be gradually raised (see NOTE in 4.6.1) to +150°F. The body assembly shall remain at this temperature for a period of 5 hours. At the completion of this 5 hour period and while at +150°F, the body assembly shall again be subjected to the tests in 4.6.7.16.1 to 4.6.7.16.4 inclusive, to determine compliance with the applicable requirements of 3.7.1.13.1 to 3.7.1.13.4 inclusive, for this temperature (+150°F). Upon completion of this test, the temperature of the test chamber shall be gradually reduced (see NOTE in 4.6.1) to room ambient temperature (+60° to +90°F) and the body assembly removed from the test chamber. Body assemblies which have passed the above tests shall be subjected to the test in 4.6.3.

4.6.3 Collimation, pre-vibration setting.- Prior to the vibration test of 4.6.4, the body assembly shall be positioned as in 4.6.7 to insure conformance to 3.7.1.2.1. While viewing through the body assembly eye-piece, obtain coincidence of the body assembly reticle boresight cross and the projected target image crosslines, by means of the boresight knobs. When coincidence has been obtained, slip the scales of the boresight knobs to a setting of (4) and recheck to insure coincidence is not disturbed.

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Record these settings for use in the test in 4.6.7.1. Without disturbing the boresight knob scale settings of (4), the body assembly shall then be subjected to the test in 4.6.4.

NOTE: Insure that knob scales are not distrubed (re-slipped) from their original setting of (4) throughout remainder of tests.

4.6.4 Vibration. - This test shall be performed on body assemblies procured as a repair part item. When the body assembly is provided with a complete periscope this vibration test will not be required when the complete periscope is to be vibrated. Standard vibrating equipment capable of providing the amplitude and frequency specified in 3.4.3 shall be utilized for this test. Unless otherwise specified, the test equipment shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The body assembly shall be clamped on the vibrating equipment in its normal mounting or operating position (as depicted in the front view on Sheet 1 of Drawing F8619880) by means of an appropriate adapter. The vibration motion shall be in the vertical plane for the time duration specified in 3.4.3. Immediately following the vibration cycle, and prior to the examination for physical defects resulting from poor workmanship and materials, the body assembly shall be subjected to the test in 4.6.7.7. Do not disturb the boresight knob scale settings of (4) during transfer of the body assembly. Subsequent to the test in 4.6.7.7, the body assembly shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.4.3, prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. With the exception of the slip scales, a tactile examination shall be made for functioning of moving parts (boresight knobs, boresight knob scales, etc.) for undue irregularities of movement, friction or looseness. An audible examination shall also be made of the body assembly for loose internal optical and mechanical components. Any one of the possible defects defined above shall be cause to consider a body assembly defective. Body assemblies which have passed the above examinations shall be subjected to the remaining tests specified in 4.6.5 through 4.6.7.16 inclusive.

4.6.5 Cleanliness - The optical surfaces of the body assembly shall be inspected for cleanliness in accordance with MIL-O-13830. The body assembly shall conform to the requirements of 3.5.

4.6.6 Sealing - This portion of the test shall be performed utilizing a pressure gage capable of reading between 0.25 to 0.50 pound per square inch gage (psig) and a dew point tester, The following test sequence shall be adhered to. Check the internal pressure of the body assembly utilizing the pressure gage specified herein. Then, test the dew point of the interior atmosphere of the body assembly by means of the dew point tester. Interior atmosphere of the body assembly shall conform to the requirements of 3.6. Upon satisfactory completion of this portion of the test internal pressurizing of the body assembly shall be performed with special testing equipment conforming to 4.5.2.3.1. Pressurize the body assembly with dry nitrogen gas, having a dew point at least as low as minus 25 degrees Fahrenheit, to 5 pound per square inch (psi), in accordance with the procedure for internal testing on Drawing F8565556. Retain the applied pressure in the body assembly for the time duration

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specified in 3.6. During this time period, the boresight knobs and the diopter adjustment shall be rotated to their clockwise and counterclockwise stops. At the end of this time period there shall be no evidence of leakage as indicated on the pressure gage of the testing equipment, to determine compliance with 3.6. Upon satisfactory completion of the sealing test, the body assembly shall then be flushed with the dry nitrogen gas specified herein. The interior atmosphere of the body assembly shall be periodically sampled through the dew point tester until the atmosphere shows a dew point at least as low as minus 25 degrees Fahrenheit. When this atmosphere is attained within the body assembly, the body assembly shall then be sealed. The internal atmosphere of the body assembly shall conform to the requirement of 3.6.

4.6.7 Performance - The following tests shall be performed with the body assembly mounted on special testing equipment conforming to the requirements of 4.5.2.3.2. Set-up of the test equipment shall be in accordance with the set-up instructions contained on Drawing A10531935, set-up instructions for Test Fixture, F8575858, Periscope Body, M34; Drawing F8577787, set-up instructions for Universal Test Stand and Drawing F8298789, Projector Collimating, Parabolic. Mounting and securing of the body assembly on the test equipment shall be performed in accordance with Drawing A10531936, operating instructions for Test Fixture F8575858, Periscope Body Assembly, M34. During these tests, the Collimator Projector, F8298789 shall be adjusted for 1200 meters and the infrared filter shall be removed from the collimator eyepiece. Unless otherwise specified, the body assembly eyepiece shall be set at the proper interpupillary distance to suit the individual's requirment and the diopter adjustment knobs of the left and right eyepiece shall be set for best focus of the body assembly reticle and the target image.

4.6.7.1 Collimation, post vibration - Observe through the body assembly eyepiece and adjust by means of the boresight knobs to obtain coincidence of the body assembly reticle boresight cross with the target image. Boresight knob adjustments shall not exceed the tolerance specified in 3.7.1.6 as measured on the boresight knob scales.

NOTE: Do not slip knob scales.

4.6.7.2 Parallax - View through the body assembly right eyepiece and position the intersecting point of the body assembly boresight cross coincident with the intersecting point of the vertical and horizontal target line image, by means of the boresight knobs. While viewing through the right eyepiece, parallax will be recognized by any apparent displacement of the intersecting point of the body assembly reticle boresight cross with respect to the intersecting point of the vertical and horizontal target line image when the observer's head is moved from side to side or up and down. Displacement between the body assembly reticle and target image shall not exceed the tolerance specified in 3.7.1.1.

4.6.7.3 Reticle movement, elevation - Establish coincidence of the body assembly reticle boresight cross and the projected target image cross-lines by means of the boresight knobs (do not slip knob scales). While observing the boresight cross (intersecting point), rotate the elevation knob in the clockwise direction. The boresight cross shall move as specified

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in 3.7.1.2.2.1 as observed and measured on the target. Re-establish coincidence of the reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the boresight cross, rotate the elevation knob in the counterclockwise direction. The boresight cross shall move as specified in 3.7.1.2.2.1 as observed and measured on the target.

4.6.7.3.1 Reticle movement, deflection.- Perform this test utilizing the procedure in 4.6.7.3 except, the deflection knob shall be rotated in the clockwise and counterclockwise direction. Boresight cross (intersecting point) movements shall meet the requirements specified in 3.7.1.2.2.2 as observed and measured on the target.

4.6.7.4 Knob rotation.- Perform this test in conjunction with the tests in 4.6.7.3 and 4.6.7.3.1. During rotation of each boresight knob (elevation and deflection), the boresight cross (intersecting point) shall be observed and show no evidence of irregularities (jump, creep or hang-up) in direction of travel until limited by engagement of each knob stop (clockwise and counterclockwise) to be in compliance with 3.7.1.2.3. While the boresight knobs are at each stop position, back off the knobs for at least 1/4 rotation and then again return the knobs to the stop. Perform this a minimum of 3 times. During these rotations, the knobs shall show no evidence of irregularity of movement (sponginess or friction) while approaching or reaching the stop. Contact with the stop shall be solid, such that a distinct metallic click can be heard to verify that reticle travel is being limited only by means of the knob stops and not by means of contact with adjacent components to determine overall compliance with 3.7.1.2.3.

4.6.7.5 Line of sight variation.- Perform this test in a manner similar to the test in 4.6.7.3.1. During rotation of the deflection knob, any variation in elevation in the line of sight, through the boresight cross (intersecting point), shall not exceed the values specified in 3.7.1.3.

4.6.7.6 Reticle shift.- With backlash removed for direction of travel, establish coincidence of the body assembly reticle boresight cross as in 4.6.7.3. While observing the intersecting point of the body assembly reticle boresight mark, disengage the elevation boresight knob from the lock position and proceed to rotate the knob (insure backlash is not introduced) for at least 1/4 turn, then reverse rotation of the knob (backlash included) for at least 1/4 turn. During the disengagement or reversal of the boresight knob, the line of sight through the body assembly reticle boresight mark shall not shift (jump) in a direction perpendicular to the intended motion more than the tolerance specified in 3.7.1.4. In addition to checking shift at this setting, repeat the above at the midpoint and maximum point of travel in each direction for a total of 5 checks overall. These 5 checks shall also be conducted with the deflection knob in compliance with 3.7.1.4.

4.6.7.7 Boresight backlash.- While viewing through the body assembly eyepiece, disengage and rotate the boresight knobs (insure backlash is removed for direction of travel) until a line of sight, through the intersection

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of the body assembly reticle boresight mark is positioned coincident with the intersecting point of the target image crosslines. With the boresight knobs in the lock position, slip the scales to Index 4 and record the position of the line of sight. Disengage the boresight knobs and continue rotation approximately 1/2 of a revolution then return and engage the knobs to their original lock position setting of Index 4. View through the eyepiece and record the position of the line of sight through the reticle boresight mark. Any difference between the recorded positions of the line of sight for each boresight knob is the amount of backlash. Backlash in each boresight mechanism shall not exceed the tolerance specified in 3.7.1.5, as measured on the target.

NOTE: Should overtravel occur during rotation of the boresight knobs, the test is nullified and shall be repeated.

4.6.7.8 Image tilt, seven power system.- The right and left optical systems of the body assembly shall be inspected utilizing an auxiliary telescope containing a crossline reticle pattern. The vertical reticle line of the auxiliary telescope shall be aligned plumb when in the test position on a qualified level surface. With aid of the auxiliary telescope, view through the right eyepiece of the body assembly and establish one extremity of the telescope vertical reticle line coincident to the vertical target image line. The image of the vertical target line shall be parallel to the telescope vertical reticle line within the tolerance specified in 3.7.1.7, as measured on the target. Repeat the above procedure with the auxiliary telescope positioned at the left eyepiece of the body assembly. The images formed by the two optical systems of the body assembly shall meet the requirements specified in 3.7.1.7.

4.6.7.9 Parallelism of reticle and image.- Perform this test while observing through the body assembly right eyepiece with the unaided eye. Position one extremity of the body assembly reticle, 400 meter range line, coincident to the horizontal target image line. The body assembly reticle, 400 meter range line, shall be parallel to the horizontal target line within the tolerance specified in 3.7.1.8, as measured on the target.

4.6.7.10 Dioptral adjustment.

4.6.7.10.1 Zero setting.- A calibrated dioptometer with a magnification of at least 3 power (3X) shall be utilized for measuring the accuracy of the zero diopter setting of the diopter scale. Set the dioptometer objective scale to zero and adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Position the dioptometer at the right eyepiece of the body assembly. While viewing through the dioptometer, obtain best focus of the body assembly reticle by adjusting the body assembly diopter knob. Record the diopter setting on the scale. Position the dioptometer at the left eyepiece of the body assembly. While viewing through the dioptometer, obtain best focus of the target image by adjusting the body assembly diopter knob. Record the diopter setting on the scale. Each recorded diopter scale setting (left and right eyepiece) shall indicate zero

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within the tolerance specified in 3.7.1.9.1.

4.6.7.10.2 Excursion range.- Perform this test in conjunction with the test in 4.6.7.10.1. From the best focus of the body assembly reticle established in 4.6.7.10.1, rotate the diopter adjustment knob and the body assembly right eyepiece to its clockwise stop, note the diopter setting on the scale index of the diopter adjustment knob, then rotate the diopter adjustment knob to its counterclockwise stop, note the diopter setting on the scale index of the diopter adjustment knob. Repeat the above procedure for the diopter adjustment knob of the body assembly left eyepiece except, knob rotation shall be from the best focus of the target image. Each diopter movement shall conform to the requirements of 3.7.1.9.2.

4.6.7.10.3 Diopter backlash.- Perform this test utilizing the dioptometer specified in 4.6.7.10.1. The dioptometer shall be zeroed and adjusted as specified therein. Starting from the stop position, rotate the body assembly right diopter adjustment knob clockwise to the "0" position without reversing direction of rotation. Position the dioptometer at the right eyepiece and obtain best focus of the target image by adjustment of the dioptometer diopter scale. Record the dioptometer scale reading. Rotate the body assembly right diopter adjustment knob clockwise to its stop position, then return it counterclockwise to the "0" position without overtravel. Again obtain best focus of the target image by adjustment of the dioptometer diopter scale. Any difference between the first and second dioptometer diopter scale readings is the amount of backlash. Repeat the above procedure for the body assembly left diopter adjustment knob. Each diopter movement shall not exceed the value specified in 3.7.1.9.3.

NOTE: Should overtravel occur during rotation of either diopter adjustment knob to the "0" position, the test is nullified and shall be repeated.

4.6.7.11 Resolution.- The right and left optical systems of the body assembly shall be inspected for resolution utilizing the dioptometer specified in 4.6.7.10.1. The dioptometer reticle shall be zeroed and adjusted as specified therein. Set the diopter adjustment knob of the body assembly right eyepiece to indicate zero diopter. Position the dioptometer at the body assembly right eyepiece. While viewing through the dioptometer, adjust the objective scale of the dioptometer for best focus of the resolution target image. Any change in diopter readings by adjustment of the dioptometer objective scale to obtain the detectable line structure and proper line count in all four meridians shall be recorded, as measured on the dioptometer objective scale. Repeat the above procedure with the dioptometer positioned at the body assembly left eyepiece. The difference between the recorded readings for any two meridians shall not exceed the value specified in 3.7.1.10 for each optical system.

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4.6.7.12 Binocular collimation, dipvergence.- Perform this test with the aid of a viewing double collimator positioned at the eyepiece assemblies of the body assembly. The double collimator shall contain a grid reticle pattern capable of measuring collimation of the body assembly. The lines of sight of the double collimator shall be parallel and lie in the same horizontal plane. The vertical reticle lines of the collimator shall be aligned plumb when in the test position on a qualified level surface. Set the body assembly diopter adjustment knobs to zero diopter and the interpupillary distance knob (IPD) to 65 millimeters (mm). View through the right eyepiece of the double collimator and adjust the collimator to establish coincidence of the boresight point (intersecting point of the horizontal and vertical reticle lines) of the target image with respect to the boresight point of the double collimator reticle. Observe through the left eyepiece of the double collimator and measure the distance from the boresight point of the target image to the horizontal line of the double collimator reticle. This represents dipvergence. Record the dipvergence reading. Repeat the above procedure with the IPD knob set at 58 and 72 millimeters (mm). Record the dipvergence readings. Dipvergence shall not exceed the tolerance specified in 3.7.1.11, as measured on the double collimator reticle. (See 4.6.7.13 when performing this test.)

4.6.7.13 Binocular collimation divergence.- Perform this test in conjunction with the procedure in 4.6.7.12 (binocular collimation, dipvergence). While viewing through the left eyepiece of the double collimator, the boresight point of the target image shall appear on the side of the zero vertical line of the double collimator reticle which represents divergence. Record the divergence readings. Divergence shall be within the limits specified in 3.7.1.11, as measured on the double collimator reticle.

4.6.7.14 Magnification.- The right and left optical systems of the body assembly shall be inspected for magnification utilizing a dynameter and a diaphragm with a known diameter. Position the diaphragm at the entrance window of the body assembly incident to the right objective. Set the dynameter at the right eyepiece. While viewing through the dynameter, adjust the dynameter for best focus of the diaphragm image. Measure the diameter of this image utilizing the millimeter reticle scale of the dynameter. Divide the known diameter by the diaphragm image diameter as read on the dynameter reticle scale. Record this value. Repeat the magnification test procedure above for the left optical system. Record this value. Any difference between the two recorded values shall not exceed the tolerance specified in 3.7.1.12.

4.6.7.15 Unity power system.

4.6.7.15.1 Collimation.- Perform this test with the aid of the three power (3X) Telescope, F5549108 positioned at the exit window of the unity power system. View through the eyepiece of the telescope and observe the boresight point (intersecting point of the horizontal and vertical reticle lines) of the telescope with respect to the boresight point of the target image. The boresight point of the telescope reticle shall meet the collimation requirements specified in 3.7.2.1, as measured with reference to the

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target image.

4.6.7.15.2 Image tilt.- Perform this test in accordance with the procedure specified in 4.6.7.8 except, the three power (3X) telescope positioned at the exit window of the unity power system shall be utilized. The requirements of 3.7.2.3 shall be met.

4.6.7.15.3 Spherical power and astigmatism.- Perform this test utilizing the diopmeter specified in 4.6.7.10.1. The dioptometer shall be zeroed and adjusted as specified therein. A resolving power chart on a wall target representing the angular subtense for one minute of arc and a supporting medium capable of receiving the dioptometer (as well as the body assembly to be tested) shall be a part of the test equipment. Construction of the resolving power chart shall be as specified in the "Resolution Test" of MIL-O-13830 and shall contain four line sets as shown in Figure I herein. The entire clear aperture of the exit window of the unity power system shall be scanned to insure compliance with 3.7.2.2. Place the body assembly on the supporting medium and position the dioptometer at the exit window of the unity power system. While viewing through the dioptometer, adjust the objective diopter scale of the dioptometer for best focus of the vertical meridian and then for best focus of the horizontal meridian on the resolving power chart. The diopter reading for best focus of each meridian (vertical and horizontal) shall be recorded as read on the objective diopter scale of the dioptometer. The algebraic difference between the recorded readings is the amount of astigmatism. The algebraic sum of the recorded readings divided by two is the amount of spherical power. Spherical power and astigmatism shall not exceed the value specified in 3.7.2.2 when calculated at any point in the scanning area.

4.6.7.16 Operability.

4.6.7.16.1 Diopter movement.- Standard torque measuring equipment capable of measuring the torque values specified in 3.7.1.13.1 shall be utilized for this test. An adapter conforming to Drawing C8591107 shall be affixed to the diopter adjustment knob. The running torque required to rotate each diopter adjustment knob through its full range of travel in each direction shall not exceed the value specified in 3.7.1.13.1 for the temperatures indicated.

4.6.7.16.2 Disengagement of boresight knob locks.- Standard measuring equipment capable of measuring the force values specified in 3.7.1.13.2 shall be utilized for this test. The force required to disengage each boresight knob lock from the locked position shall not exceed the value specified in 3.7.1.13.2 for the temperatures indicated.

4.6.7.16.3 Rotation of boresight knobs.- Perform this test utilizing standard measuring equipment with adapter, conforming to Drawing D8566516 affixed to the knobs. The locks when disengaged shall allow the boresight knobs to rotate through their full range of travel in each direction with a running torque not to exceed the value specified in 3.7.1.13.3 for the temperatures indicated.

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4.6.7.16.4 Torque, interpupillary distance knob.- Standard torque measuring equipment capable of measuring the torque values specified in 3.7.1.13.4 shall be utilized for this test. An adapter conforming to Drawing C8591109 shall be affixed to the interpupillary distance knob (IPD). The torque required to rotate the IPD knob through its full range of travel in each direction shall not exceed the value specified in 3.7.1.13.4 for the temperatures indicated.

4.6.7.17 Reticle illumination.- This test shall be performed in a dark area. A device consisting of a .19 ampere lamp energized by 3.0 volts shall be utilized to illuminate the body assembly reticle. The reticle markings shall be clearly defined when observed through the right eyepiece of the body assembly to determine conformance with 3.7.1.14. Should the brightness of a reticle defect be greater than the brightness of a reticle line the defect shall be cause for rejection.

4.6.7.18 Interchangeability.- This test shall be performed utilizing the maximum and minimum interchangeability gages in Drawings F11732113 and F11732114 respectively, to determine compliance with 3.2.1.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 8619880.

6. NOTES

6.1 Intended use.- The Body Assembly 8619880, is a modular unit of the Tank Commander's Periscope M34, primarily used in the fire control system of the 105mm, Gun Tank M60 Series.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Applicable stock number.
(c) Applicable packaging data sheet.
(d) Selection of applicable levels of preservation, packaging and marking.
(e) Provisions for First Article testing.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1240-A336

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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